Database Tables

Core Fixed Asset Management

- Track high-value, long-term items (e.g., computers, furniture) that are uniquely identifiable, assignable, and subject to depreciation.

1. **Assets -** The central table containing a unique record for every specific physical item (e.g., Laptop S/N 9876, Desk #123). This table is the single source of truth for all unique assets.
2. **Asset_Types** - A lookup table for general categories of assets (e.g., "IT Hardware," "Office Furniture").
3. **Manufacturers -** A lookup table listing the companies that produced the asset (e.g., Dell, Steelcase).
4. **Models -** A lookup table for the specific model number of an asset (e.g., "Latitude 5400," "Aeron Chair").
5. **Vendors** - Contact and financial information for the companies from which assets and services are purchased.
6. **Contracts -** Tracks warranties, service agreements, and lease terms tied to specific Assets.

Tracking & Assignment

- handle the relationships between assets, the people using them, and where they are physically located.

1. **Employees** - The master list of all personnel who can be assigned assets or request supplies.
2. **Departments** - A lookup table for internal company departments (e.g., "HR," "IT," "Finance").
3. **Locations** - A lookup table for all specific physical places within the office (e.g., "Cubicle 3B," "Conference Room 2," "IT Spare Stockroom").
4. **Asset_Bundles** - Defines the composition of a logical unit, linking multiple **Component Assets** to a single **Parent Asset** (e.g., links Monitor ID, Keyboard ID to a Workstation ID).
5. **Asset_Assignments** - Tracks the **current and historical assignment** of an **Asset** (or **Asset Bundle**) to the responsible **Employee**.

Lifecycle & Financial Management

- manage the financial, operational, and end-of-life processes for the fixed assets.

1. **Depreciation_Methods** - A lookup table defining the accounting rules used to calculate how the asset's value declines over time (e.g., "Straight-Line").
2. **Maintenance_History -** Records all life events of an asset, including service, repairs, failure reports, diagnostic notes, and the formal justification for disposal.
3. **Asset_Disposal** - Records the final action taken when an asset is formally retired (e.g., sold, scrapped) and the date it was taken out of service.

Office Consumable Supplies (Inventory)

- manage bulk items that are consumed quickly (e.g., pens, paper, toner) where unique tracking is not required.

1. **Supplies** - The master list of unique office supply items that are consumed (e.g., "Black Ballpoint Pen," "A4 Printer Paper").
2. **Categories** - A lookup table to group similar supplies (e.g., "Writing Instruments," "Printing Supplies").
3. **Inventory_Stock -** Tracks the current stock count, unit cost, and minimum reorder level for each Supply item across all Stock_Locations.
4. **Stock_Locations** - Where the bulk supply is physically stored before being dispensed (e.g., "Main Supply Closet").
5. **Usage_Log** - Records every instance when a quantity of a **Supply** is dispensed or taken from stock by a **Department** or **Employee**.

Processes

1. **Asset Acquisition and Onboarding**

Step	**Description**	**Primary Tables Involved**
Data Setup	Ensure all supporting data (e.g., Vendor, Manufacturer, Model, and Depreciation_Method) is pre-loaded into the system.	Vendors, Manufacturers, Models, Depreciation_Methods
Asset Registration	A unique record is created in Assets for *each individual item*, including its Serial Number, Asset Tag, Purchase Date, and Cost.	Assets
Contract Linkage	If applicable, the item's warranty or lease details are recorded in the Contracts table, linked by the Asset's ID.	Contracts, Assets
Bundling (If Applicable)	If the asset is part of a bundle (e.g., monitor, keyboard), records are created in Asset_Bundles linking the components to the single Parent Asset.	Asset_Bundles, Assets

Initial Location & Status	The asset's initial status is set to "Unassigned - Active," and its initial physical location is recorded in Assets (e.g., "IT Spare Stockroom").	Assets, Locations

2. **Asset Deployment and Tracking**

Step	Description	Primary Tables Involved
Assignment Request	A request is made to assign an asset (often a complete Asset Bundle) to a specific Employee.	Employees, Assets
Assignment Record Creation	A new entry is created in **`Asset_Assignments`** linking the **Asset ID** (or **Parent_Asset ID**) to the **Employee ID** and recording the **Assignment_Date**.	Asset_Assignments, Assets, Employees
Asset Status & Location Update	The **`Status`** field in the **`Assets`** table is updated to **"Assigned,"** and the **`Location_ID`** is updated to the employee's current location (links to **`Locations`** table).	Assets, Locations
Component Linkage Update	For bundles, the system confirms that all **Component_Asset_IDs** linked in the **`Asset_Bundles`** table are now logically assigned to the employee for tracking purposes.	`Asset_Bundles`, `Assets`
Transfer Management (Relocation)	If an asset is transferred to a new user: the existing record in **`Asset_Assignments`** is closed (by adding a **`Return_Date`**), and a new assignment record is created for the new employee.	Asset_Assignments, Assets, Employees

3. **Fixed Asset Maintenance and Incident Management**

Step	Description	Primary Tables Involved
Incident Reporting	An **Employee** or technician reports an issue with an assigned **Asset**. This triggers the process.	Assets, Employees
Warranty Check & Initial Status	The system automatically checks the **`Contracts`** table to verify warranty status. The **`Assets`** record is updated to a temporary status (e.g., "In Repair Queue").	`Contracts`, `Assets`
Disposal Justification	If the item is scrapped, the **`Maintenance_History`** log is updated to include the formal **Justification** for the disposal (e.g., "Scrap - Cost of Repair Exceeds Book Value").	`Maintenance_History`, `Depreciation_Methods`
Formal Disposal & Closure	A record is created in **`Asset_Disposal`** recording the final action. The **`Assets`** status is permanently set to **"Disposed,"** and the assignment in **`Asset_Assignments`** is closed.	`Asset_Disposal`, `Assets`, `Asset_Assignments`
Resolution Logging	If repaired, the **`Maintenance_History`** record is closed, noting the final resolution and cost. The **`Assets`** status is returned to **"Assigned."**	Maintenance_History, Assets

4. **Lifecycle and Financial Processes**

Steps	**Description**	**Primary Tables Involved**
Depreciation Calculation	The system periodically runs a calculation using the `Cost` from Assets and the formula defined in Depreciation_Methods to determine the current book value.	Assets, Depreciation_Methods
Disposal Justification	If an asset is deemed unrepairable or obsolete, the Maintenance_History log is finalized to include the formal justification for scrapping.	Maintenance_History
Asset Disposal	A record is created in **`Asset_Disposal`** recording the final action (scrapped, sold, donated). The asset's status in **`Assets`** is permanently set to **"Disposed."**	Asset_Disposal, Assets
Assignment Closure	The **`Asset_Assignments`** record for the disposed item is closed out with the date of disposal.	Asset_Assignments

5. **Consumable Supply Management**

Step	**Description**	**Primary Tables Involved**
Stock Check	The system continuously checks the `Current Count` in Inventory_Stock against the `Minimum Reorder Level`.	Inventory_Stock
Stock Disbursement	An Employee or Department requests supplies. The quantity is dispensed from a Stock_Location.	Stock_Locations, Employees, Departments
Usage Logging	A record is created in the Usage_Log detailing the item, quantity, date, and recipient (Employee/Department).	Usage_Log, Supplies, Employees, Departments

Inventory Update	The `Current Count` in the Inventory_Stock table is decreased by the quantity recorded in the Usage_Log.	Inventory_Stock

Data Dictionary

I. Core Fixed Asset Management Schema

1. Assets

Field Name	Data Type	Constraints/Notes	Description
Asset_ID	INT	Primary Key (PK), Auto-Increment	Unique system identifier for the asset.
Asset_Tag	VARCHAR(50)	Unique, Not Null	The company's unique physical barcode/tag (e.g., PC-00123).
Serial_Number	VARCHAR(100)	Unique, Not Null	Manufacturer's unique serial number (essential for warranty).
Asset_Type_ID	INT	Foreign Key (FK) -> Asset_Types.Type_ID	Links to the asset category (e.g., IT Hardware).
Model_ID	INT	Foreign Key (FK)-> Models.Model_ID	Links to the specific model details.
Manufacturer_ID	INT	Foreign Key (FK)->Manufacturers.Manufacturer_ID	Links to the company that produced the item.

Vendor_ID	INT	ForeignKey(FK)-> Vendors.Vendor_ID	Links to the supplier from whom the asset was purchased.
Status	VARCHAR(50)	Not Null	Current state (e.g., **'Assigned'**, **'Unassigned - Active'**, 'In Repair', 'Disposed').
Purchase_Date	DATE	Not Null	Date the asset was acquired.
Cost	DECIMAL(10, 2)	Not Null	Original purchase price (used for depreciation).
Location_ID	INT	Foreign Key (FK) -> Locations.Location_ID	Current physical location (e.g., an office, a spare room ,branch).
Depreciation_Method_ID	INT	Foreign Key (FK) ->Depreciation_Methods.Method_ID	Links to the accounting rule for depreciation.
Current_Book_Value	DECIMAL(10, 2)	Default: `Cost`	Calculated depreciated value of the asset.

2. **`Asset_Types`** (Asset Category Lookup)

Field Name	Data Type	Constraints/Notes	Description
Type_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the asset type.
Type_Name	VARCHAR(100)	Unique, Not Null	Name of the category (e.g., "IT Hardware").
Default_Useful_Life	INT	Not Null	Standard years of useful life for this category (used in depreciation).

3. **`Manufacturers`** (Product Maker Lookup)

Field Name	Data Type	Constraint/Notes	Description
Manufacturer_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the manufacturer.
Mfr_Name	VARCHAR(100)	Unique, Not Null	Name of the manufacturer (e.g., "Dell").

4. **`Models`** (Specific Product Details Lookup)

Field Name	Data Type	Constraints/Notes	Description
Model_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the model.
Model_name	VARCHAR(150)	Not Null	The specific model name/number (e.g., "Latitude 5400").
Manufacturer_ID	INT	Foreign Key (FK) ->Manufacturers.Mfr_ID	Links the model to its manufacturer.
Specifications	TEXT		Key technical specs (e.g., "i7 processor, 16GB RAM").

5. **`Vendors`** (Supplier and Service Provider Details)

Field Name	Data Type	Constraints/Notes	Description
Vendor_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the vendor/supplier.
Vendor_Name	VARCHAR(150)	Not Null	Name of the vendor (e.g., "CDW," "Staples").
Contact_Name	VARCHAR(100)		Primary contact person at the vendor.
Phone	VARCHAR(20)		Vendor phone number.
Email	VARCHAR(100)		Vendor email address.

6. **`Contracts`** (Warranty and Lease Tracking)

Field Name	Data Type	Constraints/Notes	Description
Contract_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the contract.
Asset_ID	INT	Foreign Key (FK) ->Assets.Asset_ID	Links the contract to the specific asset.
Contract_Type	VARCHAR(50)	Not Null	Type of agreement (e.g., "Warranty," "Lease," "Service Agreement").
`Start_Date`	DATE	Not Null	Date the contract begins.
End_Date	DATE	Not Null	Date the contract expires.
Vendor_ID	INT	Foreign Key (FK) ->Vendors.Vendor_ID	The vendor providing the contract/service.
Coverage_Details	TEXT		

Contract_File_Path	VARCHAR(255)		The URL or file path to the stored contract PDF/document.

II. Tracking & Assignment Schema

7. **`Employees`** (Personnel Master List)

Field Name	Data Type	Constraints/Notes	Description
Employee_ID	INT	Primary Key (PK), Auto-Increment	Unique identifier for the employee.
First_Name	VARCHAR(50)	Not Null	Employee's first name.
Last_Name	VARCHAR(50)	Not Null	Employee's last name.
Email	VARCHAR(100)	Unique, Not Null	Employee's corporate email address.
Department_ID	INT	Foreign Key (FK)-> Departments.Dept_ID	Links the employee to their department.
Status	VARCHAR(20)	Not Null	Employment status (e.g., 'Active', 'On Leave', 'Terminated').

8. **`Departments`** (Organizational Groupings Lookup)

Field Name	Data Type	Constraints/Notes	Description
Dept_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the department.
Dept_Name	VARCHAR(100)	Unique, Not Null	Name of the department (e.g., "Quality Assurance," "Finance").

9. **`Locations`** (Physical Location Lookup)

Field Name	Data Type	Constraints/Notes	Description
Location_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the location.
Location_Name	VARCHAR(100)	Unique, Not Null	Descriptive name of the location (e.g., "Cubicle 3B," "IT Spare Stockroom").
Building	VARCHAR(50)		Building or campus name.
Floor	VARCHAR(20)		Floor number or identifier.

10. **`Asset_Bundles`** (Defines Component Groupings)

Field Name	Data Type	Constraints/Note	Description
Bundle_Record_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for this specific bundle linkage record.
Parent_Asset_ID	INT	Foreign Key (FK)-> Assets.Asset_ID	The ID of the primary item in the bundle (e.g., the Desktop PC).
Component_Asset_ID	INT	Foreign Key (FK) ->Assets.Asset_ID	The ID of the secondary item that is part of the bundle (e.g., the Monitor, Keyboard).
Date_Bundled	DATE	Not Null	Date the components were

			grouped together.

11. **`Asset_Assignments`** (Current and Historical Ownership)

Field Name	Data Type	Constraints/Note	Description
Assignment_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the assignment record.
Asset_ID	INT	Foreign Key (FK) ->Assets.Asset_ID	The ID of the assigned asset or parent asset in a bundle.
Employee_ID	INT	Foreign Key (FK) -> Employees.Employee_ID	The ID of the employee responsible for the asset.
Assignment_Date	DATE	Not Null	Date the asset was checked out/assigned.
Return_Date	DATE		Date the asset was returned/unassigned (Null if currently assigned).
Notes	TEXT		Notes about the condition or reason for the assignment.

III. Lifecycle & Financial Management

12. **`Depreciation_Methods`** (Accounting Rules Lookup)

Field Name	Data Type	Constraints/Note	Description
Method_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the depreciation method.
Method_Name	VARCHAR(100)	Unique, Not Null	Name of the method (e.g., "Straight-Line," "Double Declining Balance").

13. **`Maintenance_History`** (Service and Incident Log)

Field Name	Data Type	Constraint/Note	Description
Log_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the maintenance event.
Asset_ID	INT	Foreign Key (FK) -> Assets.Asset_ID	The specific asset involved in the event.
Log_Date	DATETIME	Not Null	Date and time the incident or service occurred/was reported.
Incident_Type	VARCHAR(50)	Not Null	Nature of the event (e.g., "Repair," "Upgrade," **"Failure Report"**).
Description	TEXT	Not Null	Detailed description of the issue or service performed.
Cost	DECIMAL(10, 2)	Default: 0.00	Cost of service or repair (if out of warranty).
Technician_Notes	TEXT		Diagnosis and final resolution or action taken.
Resulting_Status	VARCHAR(50)		The new status of the asset (e.g., 'In Repair', 'Disposed').

14. `Asset_Disposal` (Retirement Tracking)

Field Name	Data Type	Constraint/Note	Description
Disposal_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the disposal record.
Asset_ID	INT	Foreign Key (FK) -> Assets.Asset_ID	The specific asset being retired.
Disposal_Date	DATE	Not Null	Date the asset was officially retired from service.
Disposal_Type	VARCHAR(50)	Not Null	How the asset was retired (e.g., **'Scrapped'**, 'Sold', 'Donated', 'Cannibalized').
Final_Value	DECIMAL(10, 2)		The final cash/sale value received for the asset (if sold).
Justification_Log_ID	INT	Foreign Key (FK) -> Maintenance_History.Log_ID	Links to the specific **Maintenance_History** record that justified the disposal.
Certification_Status	VARCHAR(50)		Status of security/environmental checks (e.g., 'Data Wipe Confirmed', 'E-Waste Processed').

IV. Office Consumable Supplies (Inventory)

15. **`Supplies`** (Consumable Product Master List)

Field Name	Data Type	Constraint/Note	Description

Supply_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the consumable item (e.g., ID for "Black Ballpoint Pen").
Supply_Name	VARCHAR(150)	Unique, Not Null	Full descriptive name of the item.
SKU	VARCHAR(50)	Unique	Stock Keeping Unit or item number used for ordering.
Category_ID	INT	Foreign Key (FK) -> Categories.Cat_ID	Links to the general supply category.
Unit_of_Measure	VARCHAR(20)	Not Null	How the item is tracked (e.g., "Box," "Ream," "Each," "Case").

16. **`Categories`** (Supply Grouping Lookup)

Field Name	Data Type	Constraint/Note	Description
Cat_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the supply category.
Cat_Name	VARCHAR(100)	Unique, Not Null	Name of the grouping (e.g., "Writing Instruments," "Printing Supplies").

17. **`Inventory_Stock`** (Current Stock Levels and Cost)

Field Name	Data Type	Constraint/Note	Description
Stock_Record_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for this stock record (allows for multiple storage locations if needed).

Supply_ID	INT	Foreign Key (FK) ->Supplies.Supply_ID	The consumable item being tracked.
Stock_Location_ID	INT	Foreign Key (FK) -> Stock_Locations.SL_ID	
Current_Count	INT	Not Null, Default: 0	The actual quantity of the item currently in stock at this location.
Minimum_Reorder_Level	INT	Not Null	The threshold quantity that triggers a low-stock alert.
Unit_Cost	DECIMAL(10, 2)		The average cost per unit for this supply item.

18. **`Stock_Locations`** (Supply Storage Areas)

Field Name	Data Type	Constraint/Note	Description
SL_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the supply storage location.
Location_Name	VARCHAR(100)	Unique, Not Null	Name of the storage area (e.g., "Main Supply Closet," "Floor 2 Printer Cabinet").
Contact_Person_ID	INT	Foreign Key (FK) ->Employees.Employee_ID	The employee responsible for managing this stock location.

19. **`Usage_Log`** (Consumption and Dispensation Tracking)

Field Name	Data Type	Constraint/Note	Description
Usage_ID	INT	Primary Key (PK), Auto-Increment	Unique ID for the usage event.
Supply_ID	INT	Foreign Key (FK) -> Supplies.Supply_ID	The item that was dispensed.
Quantity_Used	INT	Not Null	The quantity taken from stock.
Usage_Date	DATETIME	Not Null	Date and time the supply was taken.
Employee_ID	INT	Foreign Key (FK) ->Employees.Employee_ID	The employee who requested/took the supplies (can be null if a department bulk request).
Department_ID	INT	Foreign Key (FK) -> Departments.Dept_ID	The department that consumed the supplies.